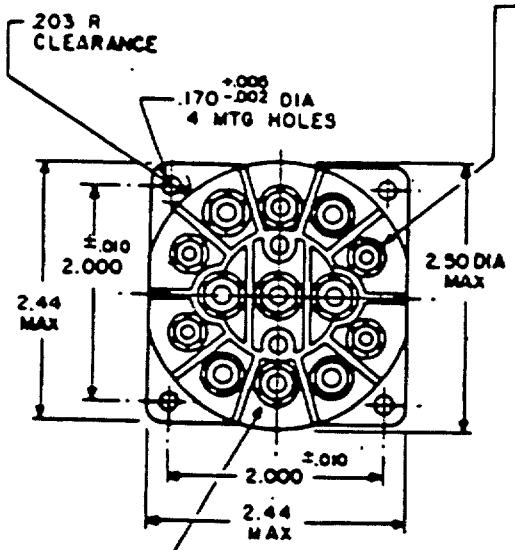
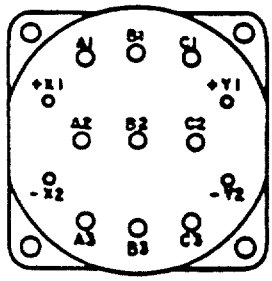
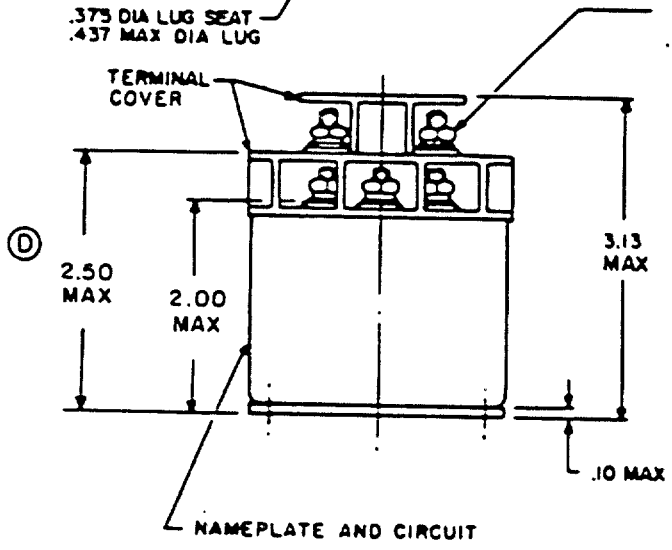
FED. SUP CLASS
5945

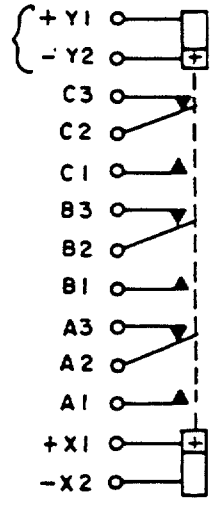




TERMINAL LAYOUT

INCHES	MM
.002	0.05
.005	0.13
.010	0.25
.10	2.5
.170	4.32
.203	5.16
.375	9.52
.437	11.10
1.91	48.5
2.000	50.80
2.44	62.0
2.45	62.2
2.50	63.5
3.13	79.5





CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.031 (0.79 mm).
3. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
4. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this standard to the extent specified herein.
5. Relay is magnetically latched in both positions.
6. Coils are not to be energized simultaneously (unless connected series aiding).
7. Metric equivalents are given for general information only.

MS PART NUMBER	TYPE	COIL	TERMINAL TYPE	MOUNTING	MAX WEIGHT IN POUNDS 1/
MS 27749-1	I	DC	STUD	PLATE	.875
MS 27749-2	I	DC	STUD	PLATE	.875

1/ Weight includes terminal barriers.

(D) Denotes change

APPROVED 17 JAN 72 REVISED (C) 19 MAR 82 (D) 29 APR 85

Review activity: AF-99

This military standard is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

P.A AF-85 Other Cust	International interest	(D) TITLE RELAYS, ELECTROMAGNETIC, 60 AMPERES, 3 PDT, MAGNETIC LATCH, HERMETICALLY SEALED	MILITARY STANDARD MS27749 (85)
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 1 OF 4

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Review activity: AF-99

DD FORM 1 MAY 73 672 (Limited coordinated) MIL-R-6106

Procurement Specification

P.A AF-85

Other Cust

International interest

SUPERSEDES:

TITLE
RELAYS ELECTROMAGNETIC, 60 AMPERES, 3 PDT, MAGNETIC LATCH, HERMETICALLY SEALED

MILITARY STANDARD

MS27749 (85)

PAGE 2 OF 4

FED. SUP CLASS 5945

OPERATING CHARACTERISTICS

Part No. MS 27749	COIL	COIL DATA									TIME-MILLISECOND-MAX					
		Nominal		Max.			Max. Pick-up Voltage			Drop Out Voltage 2/	Operate 3/	Reset 4/	Bounce			
		Volts	Freq Hz	Res Ω ±10%	Volts	Amp @ 25°C	Normal 2/	High Temp Test	Cost Current Test				Main NO	Main NC	Aux NO	Aux NC
-1	X1X2	28	DC	200	29	.190	18	20	21	-	35	--	3	3	-	-
-2 1/	Y1Y2	28	DC	200	29	.190	18	20	21 .	-	--	35	3	3	-	-

1/ Contact transfer time @ rated voltage, 1.8 milliseconds minimum.
2/ Over temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.

APPROVED 17 JAN 72 REVISED (D) FOR CHANGES SEE PAGES 1 & 2

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Procurement Specification: MIL-R-6106

P.A. AF-85

Other Cust

International interest

SUPERSEDES:

TITLE: RELAYS, ELECTROMAGNETIC, 60 AMPERES, 3 PST OR PDT, HERMETICALLY SEALED, PERMANENT MAGNET DRIVE

MILITARY STANDARD

MS27749 (85)

PAGE 3 OF 4

FED. SUP CLASS 5945

RATED CONTACT LOAD (AMPERES PER POLE) CASE GROUNDED

TYPES OF LOAD	LIFE OPERATING CYCLES x 10^3	28 V ac				115 V ac 1 Phase				115/200 V ac 3 Phase 1/				See Appropriate Notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
RESISTIVE	50	50	50			60				60				
INDUCTIVE	20	20	20											
INDUCTIVE	50					60				60				
MOTOR	50	20	20			40				40				
LAMP	50	10	10			15				15				
TRANSFER, LOAD	10					12.5/60				12.5/60				2/
MECHANICAL LIFE REDUCED CURRENT	200	7	7			14				14				
MIN CURRENT	50	1	1			5				5				

1/ Absence of value indicates relay is not rated for 3 phase applications.
2/ Transfer load indicates relay suitable for transfer between unsychronized ac power supplies at rating indicated. 12.5 amps for -1. 60 amps for -2.

APPLICATION, NOTES:
1. Examination of product, external parts shall be performed in accordance with MIL-R-6106, except the case temperature shall be limited to 150°C maximum.
2. Strength of terminals, shall be performed in accordance with MIL-R-6106, except it shall be tested at room ambient temperature only.

APPROVED 17 JAN 72 REVISED (D) FOR CHANGES SEE PAGES 1 & 2

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ENVIRONMENTAL CHARACTERISTICS

Temperature range - - - - - - - - - -55°C to +71°C
Max altitude rating- - - - - - - - - 50,000 ft
Shock G-level- - - - - - - - - - - - 50 G
  Duration - - - - - - - - - - - - - 6 ms
  Max duration contact opening - 10 us
Vibration:
  Sinusodial:
    G-level- - - - - - - - - - - - - 10 G
    Frequency range- - - - - - - - 70-2,000 Hz
  Nonoperate
    G-level- - - - - - - - - - - - - 15 G
    Frequency range- - - - - - - - 20-2,000 Hz
  Acceleration - - - - - - - - - - 15G

ELECTRICAL CHARACTERISTICS

Insulation resistance, initial - - - 100 megohms
After life or environmental tests- - 50 megohms
Dielectric strength (sea level)

	Initial	After life tests
Coil to case	1000 V rms	1000 V rms
Aux contacts		
All other points	1500 V rms	1150 V rms

Dielectric strength (altitude)

	50,000 Ft
Coil to case	500 V rms
Aux contacts	
All other points	700 V rms
Max contact drop initial	.150 volt
After life test	.175 volt

Overload current — 125 A dc; 115/200 A ac; 400 Hz 400 A
Rupture current — 150 A dc; 115/200 V ac; 400 Hz 500 A
Duty rating — Continuous

APPROVED 17 JAN 72 REVISED (0) FOR CHANGES SEE PAGES 1 & 2

Review activit AF-99

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P.A AF-85 / Other Cust	International interest	TITLE: RELAYS, ELECTROMAGNETIC, 60 AMPERES, 3 PDT, MAGNETIC LATCH, HERMETICALLY SEALED	MILITARY STANDARD MS27749 (85)
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 4 OF 4

DD FORM 1 MAY 73 672 (Linked continued)

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